Lab8 Requirements

1. 編譯第一個 driver 範例並在樹梅派上 insmod, lsmod, rmmod 指令
   - insmod

```
[root /lab8/demo/hello_module]#insmod hello.ko
[  345.212839] hello: loading out-of-tree module taints kernel.
[  345.221417] <1>I am the initial function!
[root /lab8/demo/hello_module]#
[root /lab8/demo/hello_module]#lsmod
Module                  Size  Used by
hello                  16384  0
```

   - rmmod 後執行 dmesg

```
[  345.221417] <1>I am the initial function!
[  489.859147] <1>I am the exit function!
```

2. 編譯第二個有 I/O 控制的 driver 範例與測試 driver 讀寫程式，並在樹梅派上執行 (可參考 Linux_Device_Driver_1_DOC.pdf)
   - mknod, insmod, 執行測試程式

```
[root /lab8/demo/driver_example]#mknod /dev/demo c 60 0
[root /lab8/demo/driver_example]#insmod driver.ko
[  101.671140] driver: loading out-of-tree module taints kernel.
[  101.679921] <1>DEMO: started
[root /lab8/demo/driver_example]#../
driver_example/  driver_test/     hello_module/
[root /lab8/demo/driver_example]#../driver_test/driver_test
[  168.782775] device open
[  168.787353] device ioctl
[  168.791894] device read
[  168.796309] device close
[root /lab8/demo/driver_example]#
```

▲ 若是在編譯時有遇到找不到 arm-linux-gnueabihf- (aarch64-linux-gnueabihf-) 的問題，請確認:

(1) [your path]/buildroot/output/host/bin 下的執行檔名稱，如下圖 確認 Makefile 及 make 指令的 CROSS_COMPILE 參數使用的名稱是否正確

```
jie@lab304-jie:~/Desktop/embedding_system/raspi3_32/buildroot/output/host/bin$ ls -l | grep gcc
lrwxrwxrwx 1 jie jie       17 三  17 19:07 arm-buildroot-linux-gnueabihf-gcc -> toolchain-wrapper
lrwxrwxrwx 1 jie jie       17 三  17 19:07 arm-buildroot-linux-gnueabihf-gcc-12.3.0 -> toolchain-wrapper
-rwxr-xr-x 3 jie jie  1673272 三  17 19:07 arm-buildroot-linux-gnueabihf-gcc-12.3.0.br_real
-rwxr-xr-x 1 jie jie    42608 三  17 19:07 arm-buildroot-linux-gnueabihf-gcc-ar
-rwxr-xr-x 3 jie jie  1673272 三  17 19:07 arm-buildroot-linux-gnueabihf-gcc.br_real
-rwxr-xr-x 1 jie jie    42608 三  17 19:07 arm-buildroot-linux-gnueabihf-gcc-nm
-rwxr-xr-x 1 jie jie    42608 三  17 19:07 arm-buildroot-linux-gnueabihf-gcc-ranlib
lrwxrwxrwx 1 jie jie       17 三  17 19:07 arm-linux-gcc -> toolchain-wrapper
lrwxrwxrwx 1 jie jie       17 三  17 19:07 arm-linux-gcc-12.3.0 -> toolchain-wrapper
lrwxrwxrwx 1 jie jie       48 三  17 19:07 arm-linux-gcc-12.3.0.br_real -> arm-buildroot-linux-gnueabihf-gcc-12.3.0.br_real
lrwxrwxrwx 1 jie jie       36 三  17 19:07 arm-linux-gcc-ar -> arm-buildroot-linux-gnueabihf-gcc-ar
lrwxrwxrwx 1 jie jie       41 三  17 19:07 arm-linux-gcc.br_real -> arm-buildroot-linux-gnueabihf-gcc.br_real
lrwxrwxrwx 1 jie jie       36 三  17 19:07 arm-linux-gcc-nm -> arm-buildroot-linux-gnueabihf-gcc-nm
lrwxrwxrwx 1 jie jie       40 三  17 19:07 arm-linux-gcc-ranlib -> arm-buildroot-linux-gnueabihf-gcc-ranlib
```

Lab8 Requirements

```
jie@lab304-jie:~/Desktop/embedding_system/raspi3_64/buildroot/output/host/bin$ ls -l | grep gcc
lrwxrwxrwx 1 jie jie       17 Ⅱ  18 12:00 aarch64-buildroot-linux-gnu-gcc -> toolchain-wrapper
lrwxrwxrwx 1 jie jie       17 Ⅱ  18 12:00 aarch64-buildroot-linux-gnu-gcc-12.3.0 -> toolchain-wrapper
-rwxr-xr-x 3 jie jie  1517880 Ⅱ  17 22:52 aarch64-buildroot-linux-gnu-gcc-12.3.0.br_real
-rwxr-xr-x 1 jie jie    42608 Ⅱ  17 22:52 aarch64-buildroot-linux-gnu-gcc-ar
-rwxr-xr-x 3 jie jie  1517880 Ⅱ  17 22:52 aarch64-buildroot-linux-gnu-gcc.br_real
-rwxr-xr-x 1 jie jie    42608 Ⅱ  17 22:52 aarch64-buildroot-linux-gnu-gcc-nm
-rwxr-xr-x 1 jie jie    42608 Ⅱ  17 22:52 aarch64-buildroot-linux-gnu-gcc-ranlib
lrwxrwxrwx 1 jie jie       17 Ⅱ  18 12:00 aarch64-linux-gcc -> toolchain-wrapper
lrwxrwxrwx 1 jie jie       17 Ⅱ  18 12:00 aarch64-linux-gcc-12.3.0 -> toolchain-wrapper
lrwxrwxrwx 1 jie jie       46 Ⅱ  18 12:00 aarch64-linux-gcc-12.3.0.br_real -> aarch64-buildroot-linux-gnu-gcc-12.3.0.br_real
lrwxrwxrwx 1 jie jie       34 Ⅱ  18 12:00 aarch64-linux-gcc-ar -> aarch64-buildroot-linux-gnu-gcc-ar
lrwxrwxrwx 1 jie jie       39 Ⅱ  18 12:00 aarch64-linux-gcc.br_real -> aarch64-buildroot-linux-gnu-gcc.br_real
lrwxrwxrwx 1 jie jie       34 Ⅱ  18 12:00 aarch64-linux-gcc-nm -> aarch64-buildroot-linux-gnu-gcc-nm
lrwxrwxrwx 1 jie jie       38 Ⅱ  18 12:00 aarch64-linux-gcc-ranlib -> aarch64-buildroot-linux-gnu-gcc-ranlib
```

```
jie@lab304-jie:~/Desktop/embedding_system/lab8/64/hello_module$ make ARCH=arm64 CROSS_COMPILE=aarch64-buildroot-linux-gnu-
make -C "/home/jie/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom/" M=/home/jie/Desktop/embedding_system/lab8/64/hello_module modules
make[1]: Entering directory '/home/jie/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom'
make[1]: Leaving directory '/home/jie/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom'
jie@lab304-jie:~/Desktop/embedding_system/lab8/64/hello_module$
jie@lab304-jie:~/Desktop/embedding_system/lab8/64/hello_module$
jie@lab304-jie:~/Desktop/embedding_system/lab8/64/hello_module$ make ARCH=arm64 CROSS_COMPILE=aarch64-linux-
make -C "/home/jie/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom/" M=/home/jie/Desktop/embedding_system/lab8/64/hello_module modules
make[1]: Entering directory '/home/jie/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: aarch64-buildroot-linux-gnu-gcc.br_real (Buildroot 2023.11-1177-g6c455505eb) 12.3.0
  You are using:          aarch64-linux-gcc.br_real (Buildroot 2023.11-1177-g6c455505eb) 12.3.0
  CC [M]  /home/jie/Desktop/embedding_system/lab8/64/hello_module/hello.mod.o
  LD [M]  /home/jie/Desktop/embedding_system/lab8/64/hello_module/hello.ko
make[1]: Leaving directory '/home/jie/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom'
jie@lab304-jie:~/Desktop/embedding_system/lab8/64/hello_module$
```

(2) make ARCH=arm CROSS_COMPILE=arm-buildroot-linux-gnueabihf-

make 時須注意路徑，如果不是在 [your path]/buildroot/output/host/bin 執行， cross_compile 參數要用絕對路徑，或者把 [your path]/buildroot/output/host/bin 加入環境變數。

```
jie@lab304-jie:~/Desktop/embedding_system/lab8/64/hello_module$ make ARCH=arm64 CROSS_COMPILE=~/Desktop/embedding_system/raspi3_64/buildroot/output/host/bin/aarch64-linux-
make -C "/home/jie/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom/" M=/home/jie/Desktop/embedding_system/lab8/64/hello_module modules
make[1]: Entering directory '/home/jie/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: aarch64-buildroot-linux-gnu-gcc.br_real (Buildroot 2023.11-1177-g6c455505eb) 12.3.0
  You are using:          aarch64-linux-gcc.br_real (Buildroot 2023.11-1177-g6c455505eb) 12.3.0
  CC [M]  /home/jie/Desktop/embedding_system/lab8/64/hello_module/hello.mod.o
  LD [M]  /home/jie/Desktop/embedding_system/lab8/64/hello_module/hello.ko
make[1]: Leaving directory '/home/jie/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom'
```